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Churn management in hospitality

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Abstract

This research paper presents a novel churn management approach tailored to the hospitality industry. Focused on enhancing customer retention, the proposed model consists of two integral pillars: churn identification and proactive action. Leveraging the BG/NBD model as a foundation, the identification phase is innovatively extended to accommodate multiple interactions, ensuring a more accurate prediction of customer churn. The proactive strategy, utilizing a reinforcement learning framework, dynamically establishes optimal churn thresholds through analysis of responses of guests who received a targeted email with a fixed discount. A multi-week testing phase is included within a prominent North American hotel chain, yielding empirical insights into the model's effectiveness. The results indicate that guests with a churn risk of 75% or higher should be targeted with a 20% discount email. The effectiveness of sending out the email from this churn risk becomes increasingly apparent.

Keywords: Churn management, Churn risk, Churn strategy, Hospitality

Introduction

Customer retention and customer churn are opposites in the loyalty spectrum when selling a product or service. A customer churns when they decide to cease interacting with a brand or company [28]. This leads to a customer no longer buying a product or service, which results in a future income equal to zero. A company or brand increases potential future income by increasing the number of customers who actively buy the product or service, which is why customer churn should be avoided.

All active customers form the active customer base; without one, a company or brand would not exist [19]. This base has an inflow, from new customers, and an outflow, from churning customers. A company or brand stimulates this inflow via marketing, e.g., promotion, discounts, and referrals. However, the costs of attracting or gaining new customers can be five times higher than retaining one [27]. Therefore, keeping customers active is often cheaper than attracting new ones.

Churn management focuses on keeping customers active by preventing them from churning, which is the scope of this research. Churn management consists of two equally important pillars: (1) identification of customers who are at risk of churning and (2) acting to keep these at-risk customers [32]. The vast majority of churn management research focuses on identifying churning customers only, without applying an associated

strategy on how to continue with this calculated risk. There is a research gap for applying a strategy to avoid churning customers based on their estimated risk.

Research objective

The primary objective of this study is to provide insights and recommendations to hospitality organizations, aiding them in improving their churn management practices and retaining customers within the context of a highly competitive industry. The specific aim is to augment the active customer base by effectively addressing churning customers through comprehensive consideration of their interactions.

To accomplish this, a novel churn management framework is introduced, which extends an existing churn identification model and incorporates an adaptive churn strategy. This approach is particularly well-suited when no data is present about customer churn. It is based on a reinforcement learning model, enabling the system to continually learn and optimize its response to customer churn, even in data-scarce situations. A unique empirical case study in hospitality is included, which serves to test the approach's effectiveness and offer recommendations on when to trigger actions aimed at preventing customer churn.

This paper is organized as follows: an overview of background information is described in Sect. 2, the methodology is described in Sect. 3, the empirical case study is covered in Sect. 4, and the conclusion and discussion are presented in Sect. 5.

Background

An active customer interacts with a brand or company in a set period of time [33]. As time elapses without any interaction, the likelihood of the customer becoming inactive rises [16]. Via customer engagement strategies, companies invest in interactions with customers through various channels to strengthen or activate this relationship [35]. The objective of this investment is to increase the cumulative profitability of a customer. This interaction results in a pattern of timestamped data points.

The collection of timestamped data points forms a timeline of customer behavior [13]. This timeline includes various behavioral aspects, such as customer visits, purchases, and interactions with marketing emails. The diverse range of data points within the timeline offers insights into the level of engagement between the customer and the respective brand or company [40]. The level of engagement of a customer can be expressed in the risk of churning. A customer who purchased a single product and visits the website bi-weekly would have a lower churn risk than a customer who only purchased a single product and not visiting the website at all. By analyzing the frequency of customer actions and the recency of their occurrences, it becomes possible to identify indicators that help assess the risk of churn [33].

Churn management focuses on deliberately churning customers [34]. This category of deliberately churning customers happens because competitors offer, in the eyes of the customer, an alluring alternative if the customer still needs such a product or service. This alternative can be expressed through higher quality of service, lower price or higher convenience [21]. Customer engagement focuses on the process of building a relationship with the customer to compete with the alluring alternative [24]. A churn strategy

is part of customer engagement which focuses on the customer at risk to churn [8]. This strategy creates an additional inflow to the active customer base from the outflow.

There are two general approaches to redirecting customers from the outflow to the active customer base: (1) proactive and (2) reactive. A proactive approach is identifying and taking action against a customer who is likely to churn compared to a reactive approach where one waits until the customer requests to stop their service [11]. Within hospitality, churn is unobserved since there is typically no subscription. Therefore, a reactive approach cannot be applied. In addition, preventing a customer from churning is easier than gaining a new one since attributes of the customer are known, such as historical interactions and contact information [18]. This research therefore focuses on a proactive approach, which requires estimating the risk of churning on a customer level [28].

Estimating churn risk is a widely studied problem with different approaches. The objective of the model is to estimate if a customer is churned, indicated as 1, or not churned, indicated as 0 [21, 25]. The greater the risk, the more likely that customers can churn. For a proactive approach, a decimal value is preferred compared to a classification since the objective of the strategy is to prevent churning. If a classifier labels a customer as churned, a churn strategy would be reactive. The variables for predicting churn originate from the timestamped behavioral information. Feature engineering may be required to transform the timestamped data points to input variables to evaluate a customer [32]. An evaluated customer base serves as input for the churn strategy.

Churn management can be understood in two distinct aspects, with the specific strategy employed depending on churn prediction. However, while existing research predominantly focuses on churn prediction itself, it provides limited information on strategies to effectively utilize these predictions. Although certain strategies are mentioned in the literature [21, 25], the rule of thumb is to target customers with the highest risk. However, focusing particularly on customers with the highest predicted risk of churning is not necessarily the most effective approach for proactive churn programs [2]. For example, it is possible to set a threshold for the risk and if a customer passes it, execute a retention strategy [21]. The response differs per strategy and this response is unknown when implementing a new strategy.

A strategy influences the response of the customer heavily; for example, the responses of customers at risk increase when the discount of a product or service increases [1]. It is essential to understand the effectiveness of the actions of a strategy [22]. If the action to a customer is a discount email, a response can be defined as an open, click or conversion with the discount. Given the set of evaluated customers based on churn risk, it is unknown how they respond to a specific action [26]. With a threshold risk as a strategy, there is a potential that the strategy is activated too early, when customers are still active, or too late, when customers are not interacting anymore. An adaptive strategy resolves this unknown response since it learns from the responses of earlier actions [7]. A closed feedback loop of actions to customers and corresponding responses is essential for a strategy with optimization incorporated.

The effects of churn management require measurement with metrics that support the business objective. A customer retention strategy [20] measures the effect by evaluating the recurring customers share, satisfaction, lifetime value, and customer age [4, 30]. The

strategy is executed by marketing departments, which are under increasing pressure to demonstrate their contribution [29]. Therefore, an investment in measurement is essential to quantify the contribution to the business objective.

Churn management is part of a personalized strategy of a company or brand [18]. Resources, such as time and budget, are more effective with a personalized approach compared to a mass-market approach [9]. Hospitality is an industry where avoiding customer churn and increasing loyalty contributes to higher revenue in the long-term [6]. Another relevant factor is digitalization, which allows for the analysis of more data points than only guest stays. Therefore, the need emerged to not only consider transactional data points but especially interaction data points, such as website logins, opening marketing emails, and surveys. Connecting these data points from a centralized IT infrastructure system is a challenging requirement for the success of churn management. This collection of timestamped data points forms a complete timeline of customer behavior, which serves as input for churn management.

Within hospitality, guests can be segmented into categories with similar behavior based on a set of attributes [37], which are derived from the timestamped data points of guests. These clusters provide an initial split of guests, which can serve as input for churn management. There are different types of guests staying in the hotel, with the most common split being business and leisure [39]. The risk evolution of these two guest types differs since they show different behavior. For example, a marketing department might want to take action if a frequent business traveller does not come after a time period of 3 to 6 months, while a marketing department will not take action after the same period of time for a yearly recurring leisure guest. Therefore, there is a requirement to consider segments to differentiate behavior when applying churn management [4].

Methodology

This section explains the methodology of the two pillars of the framework that form churn management: (1) churn identification and (2) churn strategy. The main advantage of this framework is a proactive and personalized strategy to avoid customers churning from the active customer base. From a business perspective, both pillars are transparent and white-box approaches, which significantly enhances the adoption rate of this framework for marketing departments. The churn identification model calculates a churn risk per customer profile, ranging from 0 to 1, while the churn strategy offers an optimal threshold for identifying customers necessitating proactive retention efforts.

The assumptions and requirements of the model are discussed in Sect. 3.1. The input of the model is past interactions per customer of a complete active customer base. The churn identification component, see Sect. 3.2, predicts the churn probability per customer, which serves as input for the churn strategy. An adaptive strategy, see Sect. 3.3, is formulated to deal with the unknown response rate of the proactive action to retain customers.

The motivation for this adaptive strategy becomes particularly relevant when no pre-existing churn strategy is available, leaving a void in initial data for predictive modelling. In such cases, this dynamic approach continuously learns and refines the optimal churn risk threshold for deploying proactive, tailored strategies. Its fundamental strength lies in recognizing the interrelation between churn identification and strategy. By doing so, it

adapts to evolving customer behaviors, filling the data gap and enhancing the organization's ability to proactively address churn in data-scarce environments.

Model

The timeline of data points per customer forms the input for the churn identification model. These data points represent interactions with the company or brand, such as visiting websites, opening marketing emails, and visiting the hotel. The importance of interaction differs since not all of the interactions contribute to the business objective equally. For example, opening a marketing email and visiting the hotel for a stay contribute differently to revenue-oriented business objectives. An example of a revenue-oriented business objective is to increase the average lifetime value of guests. It does not imply that data points, which generate no direct revenue, are unimportant for churn management since engagement can lead to revenue in the long term [38]. The importance of data points is a requirement of the churn identification model.

The set of active customers is defined as $C = \{c_1, \dots, c_S\}$, where $S \in \mathbb{N}^+$ represents the size of the set. Once the active customer base is evaluated by the churn identification model, every customer has a predicted churn probability, which serves as input for the churn strategy. A customer i , represented by c_i , has an associated churn probability, defined as p_i^{churn} . Over the passage of time, this churn probability dynamically evolves, requiring periodic reevaluation of the active customer base to ensure the accuracy of individual churn probability estimations. For any c_i , the churn probability at a distinct time point t is expressed as $p_{i,t}^{churn}$, where the parameter t ranges inclusively from 1 to T , representing a defined time interval.

The next step involves implementing a proactive and personalized strategy to target customers at risk of churning. In the context of this study, the strategy is defined as an email containing a predetermined discount code, a decision established by the hotel chain's marketing team. The responsibility for creating these specific emails lies with the marketing team of the company or brand. For the purpose of this study, a discount, denoted as e , on the subsequent transaction, referred to as q , is firmly set at 20% of the total value associated with the next stay. A business limitation is put in place, specifically capping the number of emails that can be dispatched via this strategy since offering a discount impacts profitability. The strategy aims to target a subset of customers, denoted as $C_t \subseteq C$, at a given time t within the range of 1 to T , where T represents a specific time frame.

It is important to recognize that guests with varying churn probabilities interact differently with this proactive and personalized email. The underlying intuition is that guests with a lower churn risk are more likely to engage compared to those with a higher churn risk, as churn risk serves as an indicator of their level of engagement [31]. This interaction or response is represented as a binary value, where 0 indicates no interaction (failure) and 1 signifies interaction (success). Interactions can be defined as opening emails, subscribing to personalized offers, or making reservations, which depend on the business objective. By deducing the response rate associated with each churn probability, denoted as p^{churn} , across a customer group, it becomes possible to identify an optimal threshold for dispatching the personalized email, which contains the discount.

Another business limitation dictates that each customer is eligible to be impacted by churn management only once. This implies that $C_j \cap C_k = \emptyset \forall j, k$ where $j \neq k$. When c_i is selected for receiving the discount, it is removed from the pool of eligible customers who can receive a discount until the customer makes a conversion. Otherwise, the customer could receive and submit multiple discounts, with a potential risk of loss in revenue.

Churn identification

The churn identification model designed for the hospitality sector must adhere to a specific set of requirements. A crucial requirement arises from the nature of the hospitality industry itself. Unlike contractual businesses, the hospitality industry operates on a non-contractual and continuous basis, allowing guests to make reservations and stay at hotels whenever they need to. As a consequence, the churn identification model must address the challenge of not being able to directly observe churn events and instead estimate the likelihood of a guest returning in the future.

Several modelling approaches were considered from existing literature, but were found to be unsuitable given the specific requirements outlined in this section. Previous churn research focussed on binary classification methods, such as support vector machines and random forests, providing limited guidance for a proactive approach [21, 25]. By reducing churn to a static binary outcome, these models fail to reflect how churn risk evolves over time. Survival analysis models, such as the Cox proportional hazards approach [36], offer clear interpretations through hazard ratios but lack the flexibility needed for dynamic updates of churn predictions over time. Hidden Markov Models [3] provide intuitive insights into customer behavior via transitions between predefined states, but their reliance on fixed states makes proactive, real-time targeting more difficult. Recent deep learning approaches, such as particle-classification-optimization-based BP networks and Deep-BP-ANN [12], achieve high prediction accuracy but lack straightforward interpretability during the modelling stage, making them less practical for marketing teams. Lastly, boosting and bagging-based algorithms [23], operate as ensemble black-box models that do not produce intuitive, probability-based customer-level predictions over time. Therefore, we require a model specifically suited to non-contractual settings, allowing intuitive interpretations of churn risk and enabling proactive targeting of guests.

We will use the Beta-Geometric/Negative Binomial Distribution (BG/NBD) model, introduced in [14], to estimate the likelihood of a guest churning. This model is specifically designed for noncontractual settings, where there is no formal end to the customer relationship. This model assumes that customers, in this setting hotel guests, interact at a steady rate for a period of time, and then become inactive. Its strength lies in three key capabilities: it leverages historical transaction data to model behavior, predicts each guest's future interaction potential, and estimates the probability that a guest remains active after a given point in time. BG/NBD is widely recognized in academic and industry literature as a robust and effective approach for probabilistic churn modeling [36]. By utilizing the BG/NBD model, the probability of a guest remaining active is estimated. Subsequently, the likelihood of churn is assessed by subtracting the estimated probability of a guest remaining active from 1.

The BG/NBD model is characterized by its probabilistic nature, where the interactions are modelled using probability distributions. This approach involves a set of underlying assumptions that guide the modelling process. Due to the probabilistic nature, a set of parameters needs to be estimated based on historical data. We provide a brief overview of the parameters that require estimation in the subsequent paragraphs; for comprehensive details, please refer to [14].

For each $c_i \in C$, the model assumes that the repeat interaction rate follows a Poisson process with a constant rate λ_i as long as the customer is still active. This rate of repeat interactions is specific to each customer. This implies that the time for the next interaction, denoted as Δt , is modelled as an exponential distribution parameterized by λ_i .

At a customer base level, the heterogeneity in repeat behavior across the customer base C is described by a Gamma distribution. Each $c_i \in C$ has a different parameter λ_i , which represents their individual repeat behavior. The combination of these parameters forms a distribution of λ values for the entire customer base. The Gamma distribution is particularly suitable for modelling this variation as it can handle positive values and account for different underlying behaviors within segments of customers. It captures the frequency of interactions for the entire customer base by considering the shape parameter, r , and the scale parameter, α . The statistical approach to represent the active customer base purchasing patterns, a blend of Poisson and Gamma distributions, is commonly referred to as the Negative Binomial Distribution.

For each $c_i \in C$, the model assumes that the deactivation of a customer is modelled as a geometric process. After each interaction, a c_i decides to deactivate with probability $p_i^{\text{deactivate}}$, where $0 \leq p \leq 1$. This probability of deactivation is unique to each customer. The interaction after which a customer deactivates is distributed according to the shifted geometric distribution. This results in a customer requiring at least a single interaction.

On a customer base level, a Beta distribution models the heterogeneity in deactivation across the customer base. This distribution is bounded within a range of 0 and 1, which is required since on an individual level the deactivation is modelled with a probability. This distribution is parameterized by two positive shape parameters, a and b .

In conclusion, the parameters r, α, a, b need to be estimated. To estimate these parameters, the historical data points of the behavioral timeline of the customer set C are required. The historical interaction of a customer is summarized as (x, t_x, t_O) , where x is the number of interactions, t_x is the x^{th} interaction at time t_x , and t_O is the end of the observation period, where $O \leq T$.

The BG/NBD model is utilized across various industries, prompting the development of standardized implementations in popular programming languages like Python and R. In Python, the *lifetimes* package (v 0.11.1) provides the necessary functionality to obtain the parameters for the Gamma and Beta distributions. The implementation of *lifetimes* makes use of the BFGS algorithm to determine these parameters. The BFGS algorithm is an iterative optimization method for finding the minimum of a function by iteratively updating an approximate Hessian matrix based on gradient information [17].

Given the fitted parameters r, α, a, b and the historical interaction, the churn risk is determined [15]. The term $\delta_{x>0}$ is an indicator, where the value 1 is assigned when a customer has more than one interaction in a given period and otherwise 0 is assigned in the period $(0, t_O]$.

$$p^{\text{churn}} = 1 - \mathbb{P}(\text{Alive}|x, t_x, t_O, r, \alpha, a, b) = 1 - \left[\frac{1}{1 + \delta_{x>0} \cdot \frac{a}{b+x-1} \cdot \left(\frac{\alpha+t_O}{\alpha+t_x}\right)^{r+x}} \right] \quad (1)$$

The novelty of this methodology lies in its approach to estimating churn risk, which considers the combined effect of multiple interactions rather than focusing on a single interaction, typically represented by transactions. A set of interaction I types can be selected, denoted as $I = I_1, \dots, I_N$. As mentioned in Sect. 3.1, not each interaction is equally important. Hence, each interaction I_i is assigned a weight denoted by $\theta_{I_i} \in [0, 1]$, with the constraint that the sum of all weights $\sum_{i=1}^N \theta_{I_i} = 1$. There is a possibility that the interaction is not present for a guest, i.e., if they have never logged in on the website. The total weight is recalculated based on the registered interactions. The churn risk is a weighted sum of interactions:

$$p^{\text{churn}} = \sum_{i=1}^N \frac{\delta_{x_{I_i}>0} \cdot \theta_{I_i}}{\sum_{j=1}^N \delta_{x_{I_j}>0} \cdot \theta_{I_j}} \cdot (1 - \mathbb{P}(\text{Alive}| \text{Interaction } I_i)) \quad (2)$$

If there are groups of guests, who are visiting the hotel brand that show similar behavior, underlying parameters can be fitted to a group of guests with knowledge of similar behavior. Within hospitality, this can be a simplistic division between business and leisure guests. Business guest tends to visit the hotel with a higher frequency compared to leisure guests.

This model suggests that when guests churn, they can be both permanently gone and also keep their decisions hidden. They are considered permanently gone because once they decide to churn, they will never come back. But at the same time, they will not explicitly mention to the hotel that they will never come back again.

For the detailed implementation steps, pseudo code is provided in Appendix B.

Churn strategy

The churn strategy model allocates resources effectively after the at-risk guests are identified through the churn identification model. In this section, the optimization model is introduced, which is designed for resource allocation in churn management. The model runs for a defined time frame to allocate resources, denoted as T . For example, the model determines which guests receive discounts or personalized offers at each time step t . The objective of this model is to determine the optimal allocation of resources with the intent to minimize churn rates and maximize profitability. This is achieved by identifying the churn risk threshold at which personalized offers should be dispatched.

At each time t , a churn risk is estimated for each c_i , indicated as $p_{i,t}^{\text{churn}}$, which serves as information for the strategy, denoted as π , to make a decision. Initially, the response of guests to the churn strategy is uncertain, requiring an effective learning and optimization process. To effectively learn and optimize the decision-making process, the strategy divides guests into bins denoted as $m = 1, \dots, M$, where M represents the total number of bins. For each c_i at time t , the associated churn probability, denoted as $p_{i,t}^{\text{churn}}$, is assigned to the closest bin value. This is done by comparing the churn risk value to the boundaries of each bin and assigning the guest to the bin with the closest boundary. This binning approach is

incorporated because churn risk is a continuous variable, and learning individual responses for each churn risk value can be computationally intensive.

The strategy determines for at time t an action a for each bin m . The strategy can select either the action “send email” ($a = 1$) to reach out with a targeted email aimed at retaining their loyalty, or the action “do not send email” ($a = 0$). At an individual guest level, the action associated with the designated bin corresponds to the action for the guest. The action space on a customer base level at time t is defined as $\mathcal{A}_t = \{a_1, \dots, a_S\}$, where each action $a \in \{0, 1\}$.

At each time t , the strategy selects guests who are eligible to receive the promotional email, with the group size, denoted as E , determined by the business objective. A constraint is added to the action space at each t , where $\sum_{i=1}^S a_i \leq E$. In situations where a quick understanding of these unknown responses is necessary, the number of eligible customers is increased. Conversely, targeting only a single customer with a particular churn risk can lengthen the time it takes to ascertain the response.

The reward acquired by selecting action a at time step t for c_i is represented as $z_t(a_i)$. This reward may be acquired in one of the subsequent time steps and is not necessarily received immediately. However, it's essential to note that this reward should not be too far in the future, as the dynamics of the environment can change over time. Therefore, there is a necessity to define a limit on the timeframe within which rewards can be obtained. The reward per c_i is defined as the amount of the next transaction. When $a_i = 0$, the reward corresponds to the total amount of this transaction. However, when $a_i = 1$, the discount needs to be deducted from the total amount. The cumulative reward obtained from time step 1 to t_O can be denoted as $R_{t_O} = \sum_{t=1}^{t_O} z_t(\mathcal{A}_t)$.

The reward is determined by a random variable representing the retention linked to each action. This random variable follows a Bernoulli distribution with probability $p^{\text{retention}}$, where $0 \leq p^{\text{retention}} \leq 1$. The value 0 indicates that the guest will not make a future transaction, while the value 1 indicates that the guest will make a future transaction.

Per bin, an expected reward is calculated based on the predicted retention probability, defined as $\hat{z}_m$, see Equation (3). If a resource is not allocated, defined as $a = 0$, the expected reward for that guest is the retention probability, defined as $\hat{u}_m$, multiplied by the next transaction for a product or service, defined as q . The retention probability is the opposite of the churn probability. If a resource is allocated, defined as $a = 1$, the expected reward is the estimated response probability, defined as $\hat{v}_m$, multiplied by the next transaction for a product or service minus a fixed discount given with that action, defined as e .

$$\hat{z}_m = \begin{cases} \hat{u}_m^{\text{retention}} \cdot q & \text{if } a = 0 \\ \hat{v}_m^{\text{retention}} \cdot q \cdot (1 - e) & \text{if } a = 1 \end{cases} \quad (3)$$

An adaptive strategy is necessary to effectively manage the resources considering that guest behavior and needs change over time [2, 10]. This approach is based on a reinforcement learning model, enabling the system to continually learn and optimize its response to customer churn. The retention probability for action $a = 0$, denoted as $\hat{u}_m^{\text{retention}}$, is influenced by the estimated churn risk. On the other hand, the retention probability for action $a = 1$, denoted as $\hat{v}_m^{\text{retention}}$, depends on how guests react to this action for each bin m . Consequently, the response $\hat{v}_m^{\text{retention}}$ is updated based on the share of guests who responded to the action prior to triggering the set of customers in the next time step. By

learning how guests respond to these actions, insights are gathered that can be utilized in developing a well-informed strategy for future decisions.

The highest expected reward is chosen as the preferred action per bin. The associated action with the expected reward is mapped to each customer's associated bin of the active customer base C_t . Given maximum resources per time step t , denoted as E , the customers with the highest expected reward, where $a = 1$, are selected to receive an email with a fixed discount code.

As part of the reinforcement learning model, an exploration technique has been added to the policy. The Upper Confidence Bound (UCB) algorithm is a balance between exploration and exploitation as it obtains more knowledge of the environment over time [5]. The initial focus is primarily on exploration, where the policy prefers the least historically chosen actions, to exploitation, where the policy prefers the highest estimated reward. Equation (4) is defined as the policy to choose an action a at time t , where UCB is added as an exploration/explore technique. The average reward of bin m is denoted as $\hat{z}_m$. $N_t(m)$ represents the number of times bin m is chosen prior to time t . The constant v represents the confidence value that controls the level of exploration. When $T = 1$, the response distributions are unknown and customers are randomly chosen.

$$m_t = \arg \max_{m \in M} \left[\hat{z}_m + v \cdot \sqrt{\frac{\log t}{N_t(m)}} \right] \quad (4)$$

Once a targeted group of customers receives a marketing email with a code that gives a fixed discount e , the corresponding responses are collected. The churn probability of c_i at the time of sending the marketing email, along with the response received, is used as input to update the estimated retention rate $\hat{v}_m^{\text{retention}}$ for each bin m . The expected retention rate per action is then updated. This enables the formulation of an optimal strategy for determining the ideal timing of sending emails to customers at risk of churning.

The explicit implementation steps of the churn strategy model are illustrated through pseudo code in Appendix C."

Case study

This section presents a case study within the hospitality industry, providing a detailed analysis of the outcomes of applying the churn management methodology that was introduced in Sect. 3. The introduction of the data set is followed by explanatory data analysis, leveraging its insights to derive meaningful outcomes. The results provide an overview of the performance of the churn management framework and show its effectiveness. It is important to note that, prior to the introduction of our policy, the hotel chain did not have a dedicated churn management strategy in place.

Data

The data set originates from the Property Management System (PMS) of a hotel chain with 27 properties located in North America. A PMS registers and regulates reservations in the hospitality industry. The date range of the data set is from January 2018 until

March 2023. Personal Identifiable Information (PII) is excluded due to the General Data Protection Regulation (GDPR).¹

Within the timeframe spanning from the beginning of 2018 to the first of May 2023, the hotel chain registered reservations from approximately 7.5 million profiles, highlighting the significant size of the hotel chain. However, only a specific subset of these profiles were focused on after the hotel chain implemented a selection process. This selection entailed particular criteria, namely non-repeat guests from the start date of the dataset, the presence of at least one non-group booking, a reservation at one of the designated selected hotels, a reservation associated with a predefined set of rate plan codes, and the creation of a reservation in either 2022 or 2023. This selection process resulted in a set of 127, 418 profiles, which were utilized for further analysis and implementation of the churn management framework.

The guest timeline is constructed through reservation data. From the reservations, the booking and stay interactions are derived, defined as $I = [I_{Stays}, I_{Bookings}]$. The total number of registered interactions for 127, 418 includes 122, 869 bookings and 73, 236 stays. The difference between the total number of registered interactions and booking interactions is walk-ins to the hotel. The difference between booking interactions and stay interactions represents the no-shows and cancellations. The booking date and arrival date serve as timestamps for these interactions, allowing for a chronological view of the guest's journey.

To ensure the integrity and relevance of the data, transformation processes were undertaken, involving the exclusion of the COVID-19 pandemic period. Specifically, the time frame spanning from the first of April 2020 to the first of April 2021 was identified as the designated COVID period, during which the hospitality industry experienced significant disruptions. Consequently, in light of the substantial impact on guest behavior within this period, it was deemed appropriate to remove the corresponding data. As a consequence, the total number of registered interactions of 101, 441 profiles in the data set became 111, 418 and 68, 576 for bookings and stays, respectively.

After excluding the COVID period from the dataset, an additional adjustment was implemented to the time stamps of the data preceding the COVID period. Specifically, all transaction dates before the first of April 2020 were shifted by one year. This adjustment was made to preserve the chronological sequence of the data and avoid any temporal discontinuity or inconsistency resulting from the removal of the COVID period.

The dataset provides comprehensive summary statistics pertaining to three key attributes: frequency, age, and recency, for each unique profile ID per interaction type. The frequency attribute denotes the number of repeat bookings or stays made by a guest within the designated study period, computed as the total count of unique bookings or stays minus one. The age attribute represents the temporal duration from a guest's initial booking or stay until the culmination of the study period. Finally, the recency attribute characterizes the temporal span between a guest's inaugural interaction and their most recent one. This information is valuable for conducting a thorough analysis and understanding guest behavior patterns.

¹ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) (Text with EEA relevance), [2016] OJ L 119/1.

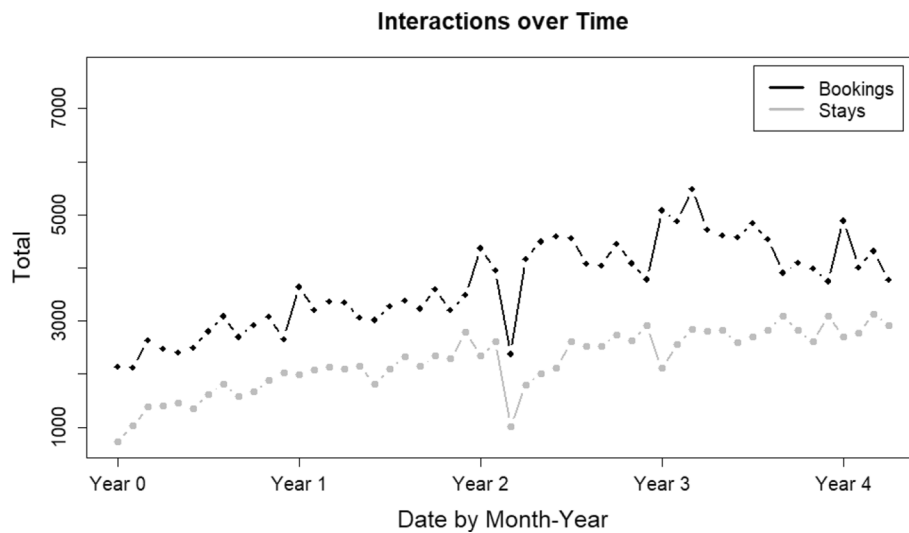


Fig. 1 Number of Transactions by Type of Interactions Over Period of Study

Explanatory data analysis

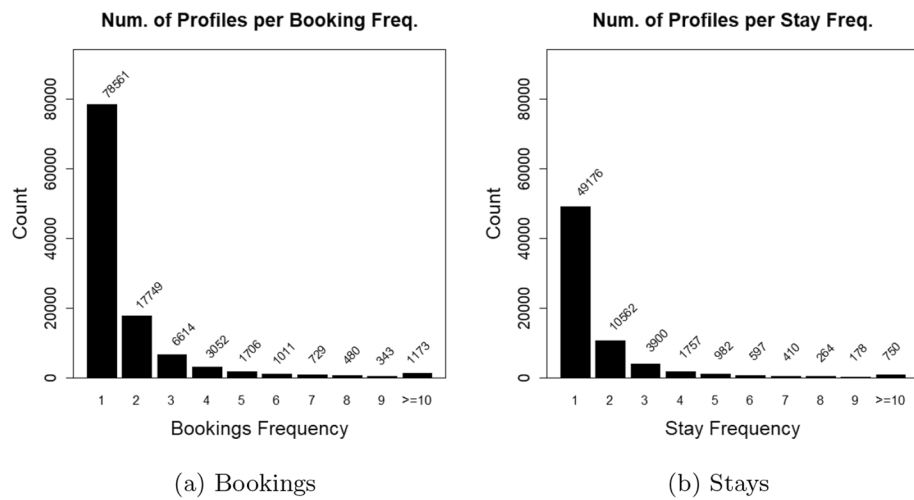
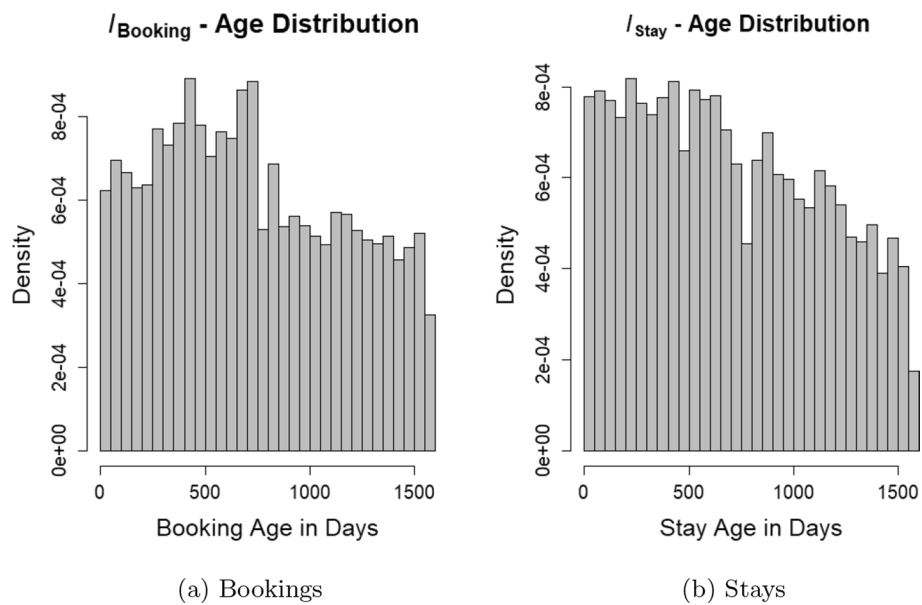
This section explores the essential profile attributes for assessing guests' churn risk, specifically frequency, profile age, and recency of different interactions. The analysis provides information about the profiles that are selected as the active customer base. Section 4.1 provides a comprehensive overview of the profile section criteria.

Figure 1 shows a monthly aggregation of the number of bookings and number of stays. The data reveal a consistent upward trend, indicating that the active customer base for this particular section of profiles is expanding.

Figure 2 shows that a substantial majority of 70% of profiles had made only one booking or stay at the properties. This observation implies that the remaining 30% of guests can be classified as repeat guests who have engaged in multiple bookings. Of these repeat guests, about 45% were observed to have stayed for a third time. Furthermore, the presence of guests who have made 10 or more bookings is notably scarce, and they have been grouped for visualization purposes, comprising approximately 1% of the active customer base.

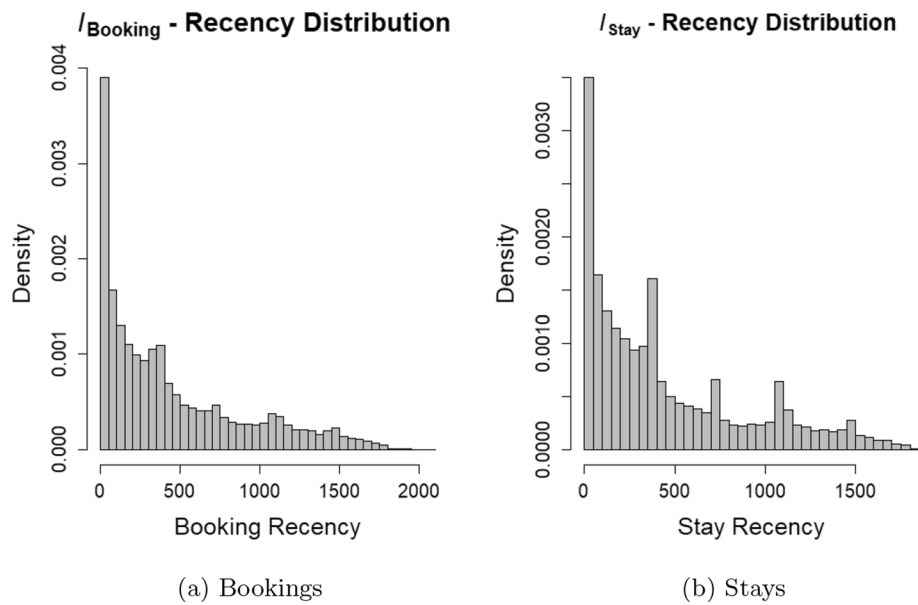
The duration, expressed in days, from the profile creation date to the end of the period under study, is denoted as the *age*. Figure 3 shows the density of profiles per interaction. Both interaction times exhibit a comparable pattern in their density plots, characterized by a downward trend as the profile's age increases. This observation can be explained by churn, wherein fewer profiles with greater ages engage over time.

The *recency* captures the temporal gap, expressed in days, between the initial and most recent interactions made by a profile. Notably, the majority of profiles exhibit a non-repeat guest behavior, as shown in Fig. 2. For the purpose of visualization, Fig. 4 only includes repeat guests. Both density plots exhibit a decaying curve shape. However, several peaks are observed at approximately 365/730/1095 days, indicating yearly repeat behavior of guests.

**Fig. 2** Interaction frequency**Fig. 3** Age density of the active customer base

Implementation

This section focuses on the implementation and configuration of the proposed framework for churn management. Initially, the division of guests into distinct clusters based on similar behavior is discussed. These clusters serve as the basis for fitting the parameters of the churn identification model. Subsequently, the settings adopted for the churn strategy are presented. Finally, the results obtained from a 6-week testing period conducted in the hotel chain's production environment are provided.

**Fig. 4** Recency density of repeat guests**Table 1** Clustering result

Cluster	Label	Size
A	Business Travelers	16.35%
B	Weekenders	31.95%
C	Early Bird - High Spenders	11.13%
D	Early Bird - Loyalist	15.28%
E	Last Minute - High Spenders	15.75%
F	Indirect - Low Spenders	5.74%
G	Loyalist - Low Spenders	3.80%

Clustering

As mentioned in Sect. 2, hotel guests visit for various purposes, and their behavior can be grouped based on the guest profile's features. The basis of these features originates from the Property Management System, and an unsupervised machine learning algorithm is applied to create clusters based on guest's behavior, as described in [37].

A hierarchical clustering technique is applied based on 364,809 profiles with 19 features, resulting in the formation of 7 clusters, which are labelled with the letters A to G. Table 1 shows a suggested name label originating from the marketing department of the hotel company, along with the cluster size represented as a percentage of the total number of guests. For simplicity, the guest profiles are only evaluated once and assigned to a single cluster before entering the churn management strategy, implying that guests do not change clusters over the testing phase.

Table 5 in Appendix A presents a summary of feature averages per cluster. The overall behavior of a cluster can be derived from guests who are assigned to a cluster based on a combination of a set of attributes. For instance, clusters D and G are composed of frequently returning guests, but their overall behavior is also influenced by other factors,

such as booking via a direct channel and the lead time (the time between the booking date and arrival date). Another interesting finding is cluster B, which shows a high percentage of short stays during weekends and a preference for booking via direct channels. The labels assigned to the clusters are created by considering three or more attributes that differentiate their behavior from that of other clusters. This helps to categorize and distinguish the clusters based on meaningful characteristics.

This outcome of the clustering divides the active customer base into separate groups, where eventually the parameters of the churn identification model are fitted. This approach allows cluster-specific churn risk estimation models, improving predictive accuracy by accounting for the unique patterns within each cluster.

Churn identification model

As discussed in 3.2, the churn identification model estimates churn probability at a particular timestamp based on the guest's historical timeline of interactions. These interactions are summarized as (x, t_x, t_O) per type, where x is the number of interactions accumulated by guests until the end of the period of study, or frequency. Meanwhile, t_x is the duration between the first stay of a guest and their latest stay, and t_O is the duration between the first stay of a guest and the end of the period under study.

There are several ways a guest can interact with hotels, from an online search on a search engine to making a booking or staying in a hotel. All these interactions are influential in describing the degree of engagement of guests and hence directly impact the churn risk. For this research, the set of interactions contains two different types of interaction, defined as $I = [I_{Stays}, I_{Bookings}]$, which have been implemented to evaluate the churn risk of a profile. As per business logic, a stay is considered more influential in describing the probability that a profile will churn than a booking itself since with a stay the guest actually arrived at the hotel. As a consequence, the weights given to these interactions are $\theta_I = [.7, .3]$, implying that a stay as an interaction will have more impact on a profile's churn risk than a booking.

Table 2 displays the fitted parameters for each cluster-interaction combination in the churn identification models. These parameters offer valuable insights into the behavior of guests before they churn in various clusters. Notably, these parameters are fitted based on the entire customer base, denoted as *Overall* in Table 2, eliminating, from a practical standpoint, the need for a cluster assignment as a prerequisite within this framework.

Concerning stay interaction, a lower parameter value r in cluster D compared to cluster E indicates a higher degree of heterogeneity in transaction rates λ among guests within cluster D. This suggests that in cluster D, there is more variation in the frequency of stays compared to cluster E, where guests may exhibit slightly more consistent stay behavior. Similarly, regarding the parameter α , cluster E displays a higher value, indicating potentially less variability in churning behavior among hotel guests. In contrast, cluster D exhibits a lower α value, suggesting a greater degree of variability in customer churn rates within this cluster.

The other fitted parameters, a and b , represent the beta distribution to model the likelihood of guests either remaining "active" or "not churning" at a given moment. Notably,

Table 2 Parameters of the churn identification model per cluster per interaction

Clusters	Interaction - Stay				Interaction - Booking			
	r	α	a	b	r	α	a	b
A	0.234	61.405	0.037	0.042	0.257	93.939	0.033	0.044
B	0.205	39.020	0.052	0.063	0.258	106.111	0.032	0.053
C	0.247	82.713	0.034	0.072	0.266	117.938	0.024	0.061
D	0.256	86.670	0.013	0.057	0.259	105.190	0.011	0.053
E	0.301	169.67	5e-13	3e-06	0.305	182.562	6e-13	3e-06
F	0.203	35.169	0.062	0.030	0.248	80.568	0.052	0.028
G	0.231	50.424	0.050	0.063	0.258	83.353	0.041	0.060
Overall	0.235	60.906	0.042	0.059	0.265	106.121	0.031	0.052

cluster D features have higher values for both of these parameters compared to cluster E. This implies that stayers in cluster D are more inclined to remain active for extended periods, while those in cluster E are likely to churn sooner.

In summary, the parameters of cluster D suggest a higher degree of variability in behavior with respect to interactions and churn risks compared to cluster E. Guests in cluster D may exhibit more diverse staying habits, greater variation in churn behavior, and an increased likelihood of remaining active for longer duration when contrasted with those in cluster E. This example underscores the concept that churn behavior varies among clusters.

To further demonstrate the aforementioned concept, a timeline is simulated that showcases bookings and stays for two guests with identical historical timelines. One guest, identified by ID 18306090, remains unassigned to any cluster, while the other guest, with ID 23101995, is assigned to cluster D. The visualization of churn risk evolution can be observed in Fig. 5. Both guests made a booking, represented by a square symbol, and a stay, represented by a triangle symbol, at precisely the same time. Notably, the churn risk is expected to be lower for the guests in cluster D, as this cluster has a higher repeat rate.

Churn strategy model

Like the churn identification process, the churn strategy model requires configuration to target or activate guests who will receive marketing emails effectively. On a weekly basis, during each batch, $E = 1,000$ guests are selected to receive an email with a discount. Within this group of size E , as an additional exploration technique besides Upper Confidence Bound, 2% of guests are randomly suggested to obtain responses over different churn bins. These selected guests receive a churn strategy email containing a 20% discount code, which they can use for their next transaction q .

To simplify matters, the variability in the next transaction for each client, influenced by the use of discount codes on various dates with different underlying rates, has been standardized by setting the next transaction parameter q to 1 within Equation (3). Nevertheless, the model retains the capability to adapt this parameter dynamically if needed.

The reward function used in this study, referred to as Equation (3), is inspired by the cost-profit dynamics framework [28], which balances the benefits and costs of retention

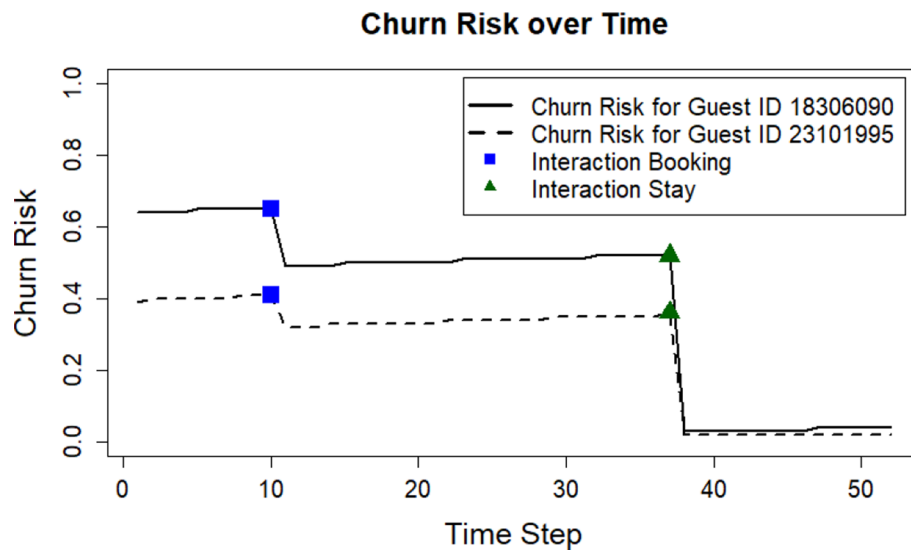


Fig. 5 Example of churn risk for an individual guest over time

strategies. This study extends its approach by embedding it within a reinforcement learning framework to optimize retention decisions dynamically. Unlike static frameworks, the reinforcement learning model leverages this cost-profit logic iteratively, adapting to evolving customer behavior and response probabilities over time. This integration enables the model to align theoretical insights with the practical needs of the hospitality industry, where customer interactions and churn risks fluctuate.

With this configuration, the reinforcement learning model gathers valuable data on the responses to the marketing message for guests with varying churn risks that are assigned to a bin m . This learning process helps the model gain knowledge about the environment over time, enabling it to optimize the marketing strategy for retaining customers effectively.

Before each trigger, the retention probabilities of Equation (3) are updated, as described in Sect. 3.3, to update the expected reward function. A time frame is set for an action for each guest to assess the effectiveness. Any booking made by a guest within the first four weeks of the time of an action is considered, and has an impact on the expected reward function.

During the first batch of marketing emails that are outgoing to guests, denoted as $t = 1$, the strategy employs a randomized approach, distributing emails across various churn bins. However, as the model continuously refines its strategy, it incorporates guest responses from prior email batches to enhance its decision-making. This iterative process ensures the selection of the most appropriate churn category when dispatching marketing emails, taking into account their latest churn probability.

The reinforcement learning framework leverages the calibrated probabilities dynamically. Customer responses to retention actions (e.g., targeted emails) are fed back into the framework, allowing for continuous refinement of the probability estimates. This feedback loop ensures that the model adapts to evolving customer behaviors over time, improving the alignment between predicted probabilities and observed outcomes. By

Table 3 Average churn risk per cluster

Cluster	Label	Average churn risk
A	Business Travelers	0.29
B	Weekenders	0.37
C	Early Bird - High Spenders	0.22
D	Early Bird - Loyalist	0.16
E	Last Minute - High Spenders	0.26
F	Indirect - Low Spenders	0.47
G	Loyalist - Low Spenders	0.28
Overall		0.29

integrating initial calibration with continuous adjustments, the proposed framework achieves a balance between theoretical robustness and practical relevance for real-world churn management scenarios.

In summary of the implementation, the NG/NBD algorithm was applied to estimate the probability of guest churn, referred to in this implementation as the churn identification model. This model underwent training and evaluation using the entire active customer base, resulting in churn probabilities for all profiles. These probabilities constitute the foundation for the churn strategy, a reinforcement learning model tasked with pinpointing the optimal profiles for targeted marketing emails.

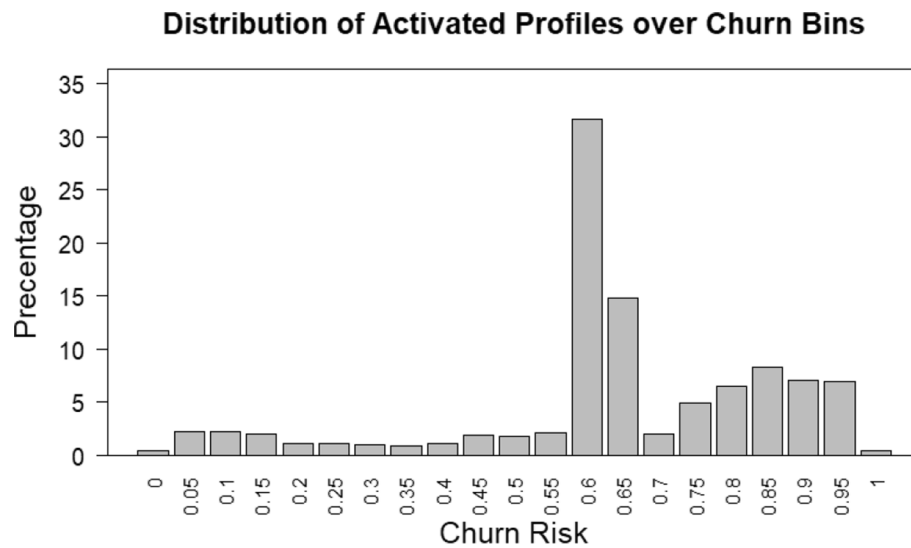
Results

A 6-week testing phase evaluated the proposed churn management strategy in the hospitality industry, combining churn estimation and reinforcement learning to determine the optimal churn risk threshold. By accurately identifying guests at risk of churning, personalized marketing interventions were created using an adaptive threshold. The dynamic nature of the reinforcement learning algorithm ensured continued effectiveness amid changing guest behaviors. These results showcase the effectiveness of implementing a data-driven churn management strategy, enabling hospitality companies to enhance customer retention and gain a competitive edge.

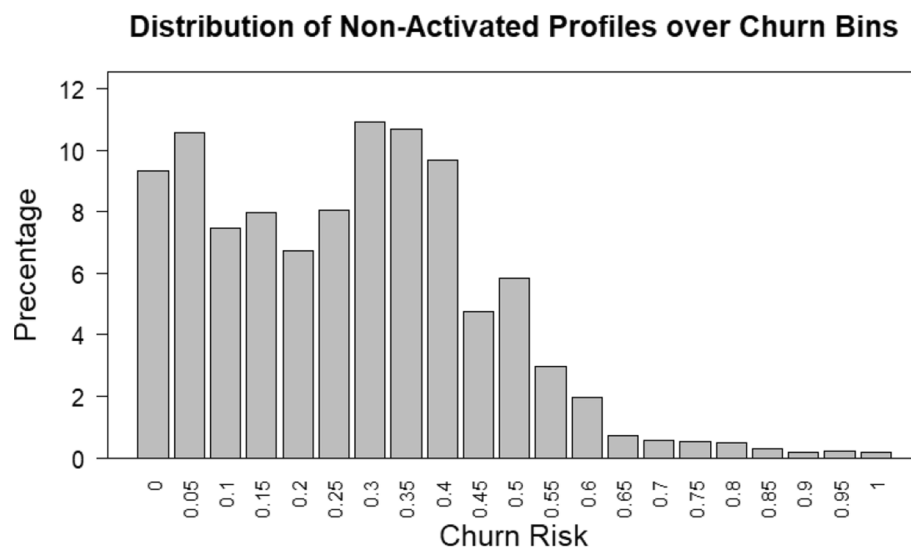
Figure 3 shows information about the average churn risk per cluster. This risk aligns with the marketing department's cluster labels, confirming their accuracy. Notably, in Cluster D, characterized by loyal customers who frequently return, the average churn risk is notably low at just 0.16. On the other hand, in Cluster F, where guests tend to stay only once with a low repeat percentage, the churn risk is high at 0.47, further validating the behavioral patterns exhibited by the guests in this cluster.

During the testing phase, a total of 7,518 profiles received an email containing a discount code, out of an eligible guest pool of 364,809. Figure 6a displays the distribution of profiles activated per churn risk bin at the end of the testing phase. This visualization shows that most activations took place for guests who are assigned to the churn bin 0.6. Figure 6b displays the distribution of the non-activated profiles of the churn bins.

The effectiveness of the churn management email campaign is examined in comparison to six other standard email campaigns that were distributed to the same hotel chain over the past six months before the testing period. Table 4 reports the campaign



(a) Activated profiles



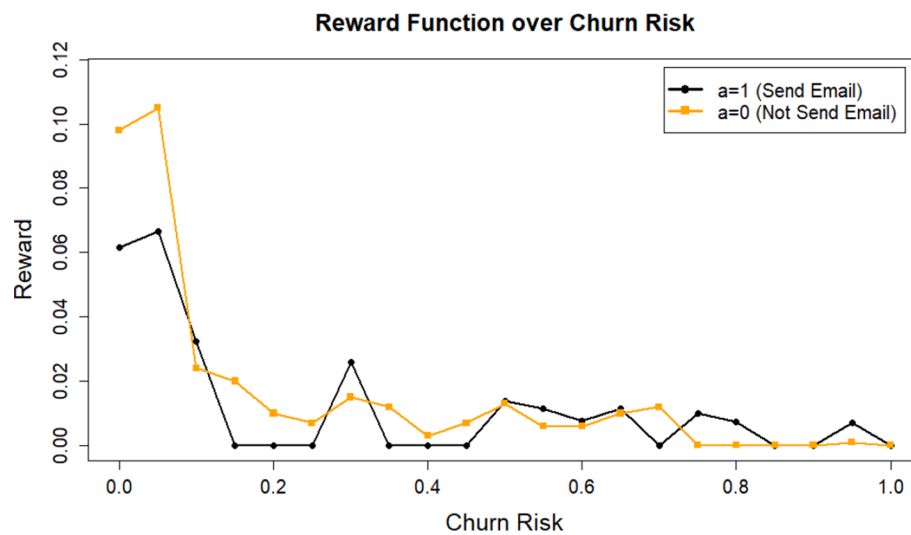
(b) Non-activated profiles

Fig. 6 Churn bin distributions

statistics in terms of audience size, opens, clicks, and conversions. The six campaigns are identified using randomly generated names. The campaigns selected for comparison were chosen because of similar incentives, such as percentage-based discounts on future bookings, and targeted similar customer segments within the same time frame of the execution of the testing phase. This selection ensures that the comparison isolates the impact of our proposed churn management strategy from other variables, providing a fair and meaningful benchmark against standard marketing practices. Additionally, measures were taken to ensure that no customer profile received more

Table 4 Campaign statistics

Campaign Name	Size	Opens	Clicks	Conversions
5651925c	13,076	3,492	242	15
c7a68444	4,620	1,226	124	12
f0400ac8	16,554	4,911	478	32
19f936f5	2,685	776	80	1
a412436d	17,080	5,534	699	32
9956178f	9,972	2,664	425	28
Churn Management	7,518	3,486	239	36

**Fig. 7** The reward function results after a multi-week testing period

than one email during the testing phase. On average, the conversion rate for these campaigns stands at 0.19%, while the churn management campaign achieves a higher conversion rate of 0.48%.

To delve into the results of the churn management strategy, the two actions are compared in Fig. 7. This figure displays the reward function values for both actions for different churn risk bins. The reward function for $a = 0$ is derived from responses of 357,291 profiles, and the reward function for $a = 1$ is based on 7,518 profiles. The reward function is based on the conversions being within four weeks of receiving the emails, as mentioned in Sect. 4.3.

Within the figure, it's evident that not all churn risk bins led to conversions during the designated testing period for sending emails. Specifically, churn risk bins 0.15, 0.4, and 0.9 did not yield conversions despite being triggered. In cases of low churn risk, the policy tends to opt for not sending emails ($a = 0$) since the rewards for this action consistently outperform those of sending emails ($a = 1$).

Beginning from the churn bin 0.5, sending emails ($a = 1$) consistently proves to be the more advantageous choice compared to not sending them ($a = 0$), except for the churn bin 0.7. This indicates that the strategy has a positive impact on guests in the associated churn bins. The retention rate nearly drops to zero for guests in churn bin 0.75 who did not receive emails, resulting in the reward function for this action

dropping to zero. The effectiveness of sending emails becomes increasingly apparent from this point onward.

Conclusion & discussion

The novel churn management approach presented in this research paper holds significant promise for any industry where multiple interactions are considered as customer engagement. This approach is demonstrated by the empirical validation conducted within a prominent North American hotel chain. Expanding on the insights obtained from a multi-week testing phase, this section provides a summary analysis, explores the implications of the findings, and lists potential pathways for novel research.

The multi-week testing phase provided validation for the proposed churn management model. The implementation of the identification step, combined with a reinforcement learning-based proactive strategy showed promising results. Based on the result section, the advice to the hotel chain is to target guests with a 20% discount email if they have a churn risk of 75% or higher. This is because the effectiveness of sending out the email becomes increasingly apparent from this churn risk onward. The adaptation of the BG/NBD model to include various types of interactions introduced a novel perspective on predicting customer churn, emphasizing the significance of considering various customer behaviors.

Building on this, our study adopts the BG/NBD model to estimate churn probability in a noncontractual setting, such as hospitality. In contrast to traditional churn prediction models that frame churn as a binary outcome, BG/NBD provides a probabilistic and time-sensitive view of guest behavior. It leverages historical transaction data to estimate the likelihood that a guest remains active over time. This approach makes it particularly suited for environments where the end of the customer relationship is unobserved. Our use of BG/NBD is well-grounded in prior research that advocates for stochastic models in customer lifetime value and retention forecasting.

This study aligns with and advances existing research efforts that aim to bridge churn modelling with actionable, strategy-oriented decision-making. While prior work, as support vector machines, random forests, and deep learning models [21, 25, 32], deliver strong predictive performance, they often lack interpretability and do not provide a dynamic view of churn risk. This limits adaptive capabilities in proactive decision-making, particularly in industries like hospitality. Our findings align with literature to move beyond static churn classification and toward methods that integrate the dynamicness of customer behavior [2, 8], enabling more precise and actionable insights.

Finally, this study advances the literature by demonstrating how churn probability modelling can be operationalized within a decision-making framework tailored to hospitality. While several researchers have proposed adaptive strategies for customer engagement [18, 24], there remains a notable lack of empirical work that ties probabilistic churn estimation to actual retention interventions in this domain. By integrating the BG/NBD model with reinforcement learning for adaptive strategy optimization, this study bridges highlights the connection between interpretability and strategic actionability. This integration lays a practical foundation for personalized, cost-efficient retention strategies that are both data-driven and operationally scalable.

In terms of future research could include refining the identification phase by incorporating additional types of guest interactions with additional interaction types. It is important to note that adding more interaction types can complicate the model, requiring thoughtful design and feature adjustments to handle the increased complexity efficiently.

Another potential focus lies in optimizing the reinforcement learning framework to adapt to a range of discount structures and personalized incentives, thereby extending the potential impact of enhancing customer retention on several levels. For instance, it could involve offering smaller discount codes to guests with lower churn risk and, as churn risk increases, providing higher-value discount codes to incentivize guests to make reservations.

While this study focuses on quantitative data, incorporating qualitative insights, such as customer feedback, satisfaction surveys, or brand sentiment, could further enrich the model's effectiveness. The framework, as outlined in Sect. 3.2 and Sect. 4.3.2, already supports diverse interaction types through adjustable weights, allowing for the inclusion of qualitative data processed via methods like sentiment analysis. Future research could explore integrating these qualitative factors to provide a more holistic understanding of churn dynamics and enable more tailored, empathetic retention strategies.

Clustering overview - attributes

Table 5 An overview of feature averages per cluster

Segment Label	A	B	C	D	E	F	G
Total Reservations	1.21	1.29	2.3	3.19	1.66	1.05	3.34
Total Stay Reservations (%)	86	86	61	79	88	63	89
Average Room Nights	1.43	1.73	3.48	2.03	3.3	2.11	1.85
Company Reservation (%)	7	8	13	29	10	15	41
Agency Reservation (%)	2	2	7	14	6	98	32
Group Reservation (%)	1	1	3	14	2	0	7
Week Stay (%)	98	2	6	39	21	60	42
Weekend Stay (%)	2	98	94	61	79	40	58
Average Lead Time	22.26	25.68	82.59	45.68	23.08	26	22.33
Reservation Source Group Direct (%)	99	98	94	86	94	1	55
Reservation Source Group In-Direct (%)	1	2	6	14	6	99	45
Total Revenue	953	1286	4926	3056	4755	802	3018
Average Revenue	819	1049	2334	1113	2225	782	1006
Repeat (%)	11	18	38	93	28	3	95
Short Stay (%)	91	85	32	81	35	72	88
Long Stay (%)	0	0	2	0	2	1	0
Medium stay (%)	8	14	66	18	63	27	12
Early Bird Booker (%)	22	26	80	47	23	26	25
Last Minute Booker (%)	54	48	7	22	39	35	39

Churn identification pseudocode

Churn Identification Model

- 1: **Input:** Interaction timelines per person (such as bookings, stays)
 - 2: **Input:** Weights for each interaction type (e.g., stays = 0.7, bookings = 0.3)

 - 3: **Process:**
 - 4: **for** each person in active base **do**
 - 5: Retrieve historical interaction timeline
 - 6: $total_risk \leftarrow 0$
 - 7: **for** each *interaction_type* in timeline **do**
 - 8: $freq \leftarrow$ count of that interaction (e.g, how many stays)
 - 9: $rec \leftarrow now - \max(timestamp \text{ of this type})$
 - 10: $age \leftarrow \max(timestamp) - \min(timestamp)$ for this type
 - 11: $(r, \alpha) \leftarrow$ fit Poisson–Gamma model
 - 12: $(a, b) \leftarrow$ fit Geometric–Beta model
 - 13: $p_{alive} \leftarrow \frac{1}{1 + \frac{a}{b+freq} \cdot \left(\frac{\alpha+age}{\alpha+rec}\right)^{r+freq}}$
 - 14: $churn_risk \leftarrow 1 - p_{alive}$
 - 15: $total_risk \leftarrow total_risk + churn_risk \cdot weights[interaction_type]$
 - 16: $final_churn \leftarrow total_risk / \sum weights[all \ types]$
 - 17: Store *final_churn* as churn risk for the current person

 - 18: **Output:**
 - 19: Churn probability score for each person
-

Churn strategy pseudocode

Churn Strategy Model

- 1: **Input:** Churn risk score per person ($p^{\text{churn}} \in [0, 1]$)
 - 2: **Input:** Fixed number of marketing actions allowed per time window (e.g., $E = 1000$)
 - 3: **Input:** Fixed discount offer to be sent (e.g., $e = 20\%$)
 - 4: **Process:**
 - 5: Initialize churn risk bins (e.g., $[0.1-0.2]$, $[0.2-0.3]$, ..., $[0.9-1.0]$)
 - 6: **for** each time window (e.g., weekly) **do**
 - 7: Assign each person to a churn bin based on p^{churn}
 - 8: **for** each bin **do**
 - 9: Estimate expected reward for each action:
 - 10: $a = 0$: do not send email
 - 11: $a = 1$: send email with discount
 - 12: Reward incorporates:
 - 13: • Estimated retention probability with and without email
 - 14: • Estimated value of next transaction
 - 15: • Cost of discount
 - 16: Apply exploration-exploitation strategy (Upper Confidence Bound):
 - 17: • Prioritize actions with high expected return
 - 18: • Explore bins with low history
 - 19: Select top E persons across bins using strategy
 - 20: Send emails with fixed discount e
 - 21: Observe responses and update:
 - 22: • Retention probabilities
 - 23: • Estimated rewards
 - 24: • Select churn risks bin for exploration based on Confidence Bound
 - 25: **Output:**
 - 26: Updated mapping of churn bins to action strategies
 - 27: Continuously improving policy for marketing interventions
-

Data availability

The data that support the findings of this study are not publicly available due to confidentiality restrictions.

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